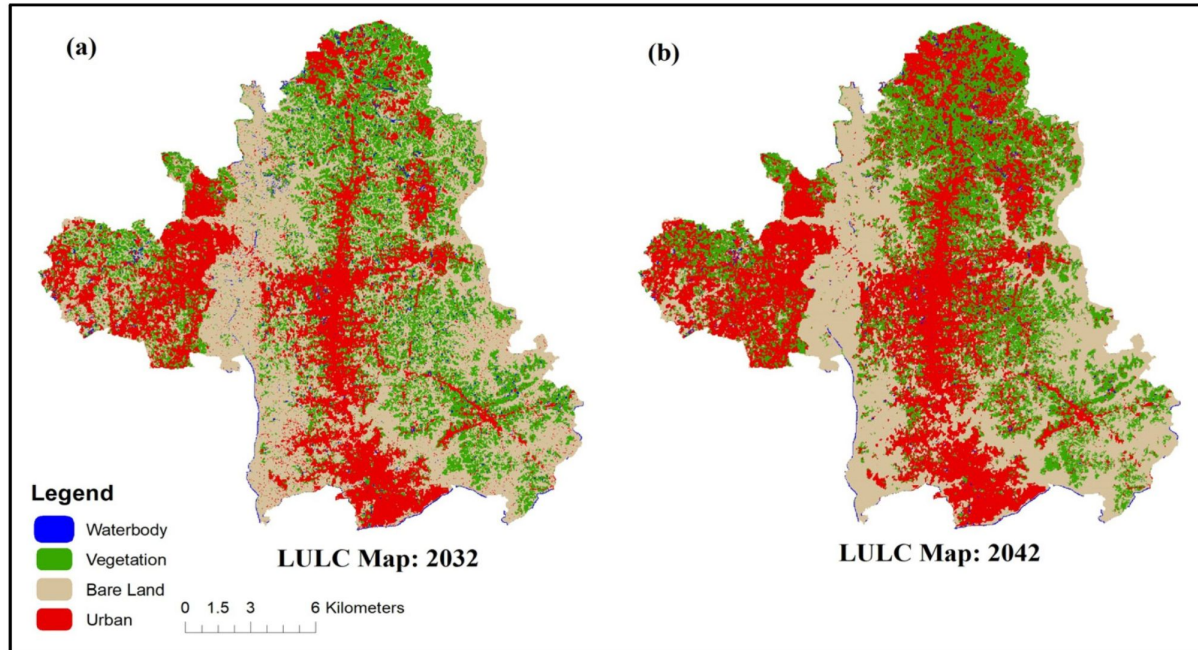


LULC Forecasting integrating Cellular Dynamics and Neural Insights



ABSTRACT

This project explores the development and application of the MOLUSCE (Modules for Land Use Change Simulations) plugin for QGIS, which assists in analyzing and predicting land use changes. Addressing the growing need for efficient land management, this project aims to provide a set of robust algorithms—including Artificial Neural Networks (ANN), Logistic Regression (LR), Weight of Evidence (WoE), and Multi-Criteria Evaluation (MCE)—to model and simulate land transformation processes. By leveraging raster data, the MOLUSCE module enables spatial analysis to predict and validate changes over time, offering tools like kappa statistics for result validation. This project emphasizes data preprocessing, training methodologies, and model evaluation, yielding significant insights for planners and researchers. Additionally, future enhancements could include integration with newer QGIS tools and improved model accuracy, contributing to sustainable land use planning efforts.

TERMINOLOGY

1. **Land Use Change:** Refers to the process through which human activities transform the natural environment, resulting in changes in how land is utilized (e.g., agricultural land becoming urban). Modeling land use change helps predict and manage future landscape transformations.
2. **QGIS (Quantum GIS):** An open-source Geographic Information System (GIS) that allows users to visualize, analyze, and interpret spatial and geographic data. QGIS supports various plugins, including MOLUSCE, which extend its capabilities for specialized applications.
3. **Raster Data:** A type of spatial data format represented as a grid of pixels, where each pixel has a value representing information like elevation or land cover.

TERMINOLOGY

4. Weight of Evidence (WoE): A method often used in spatial analysis to measure the relationship between predictor variables and outcomes. Here, it helps assess the probability of land changes based on historical data.

5. Multi-Criteria Evaluation (MCE): A decision-support tool used to evaluate and combine multiple criteria or factors. In land use planning, MCE helps integrate social, economic, and environmental factors into decision-making.

6. Kappa Statistics: A statistical measure used to validate model predictions by comparing observed and predicted results. A higher kappa value indicates better model accuracy in predicting land use change.

7. Simulation: The process of creating a model to replicate real-world scenarios. In this project, simulation helps forecast land use patterns over time based on historical data and influencing factors.

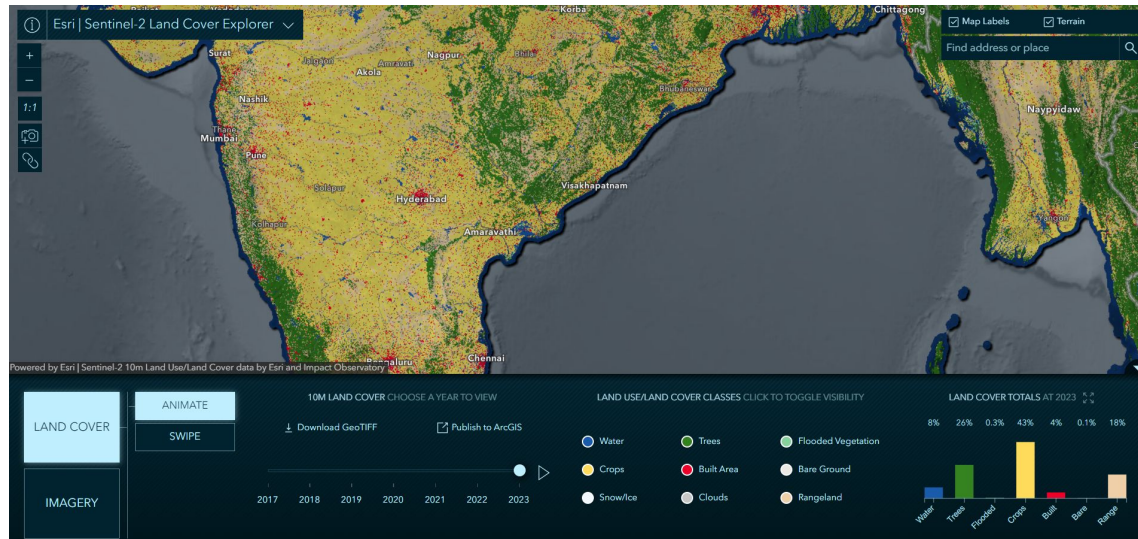
AIM OF THE PROJECT

- Predict changes in next 5 years in a specific geographic region.
- Analyze reason for the changes as Inference.
- Try using different features projected from raster data.
- Use current data from Sentinel II and experiment on different algorithms

METHODOLOGY

1. Data Collection

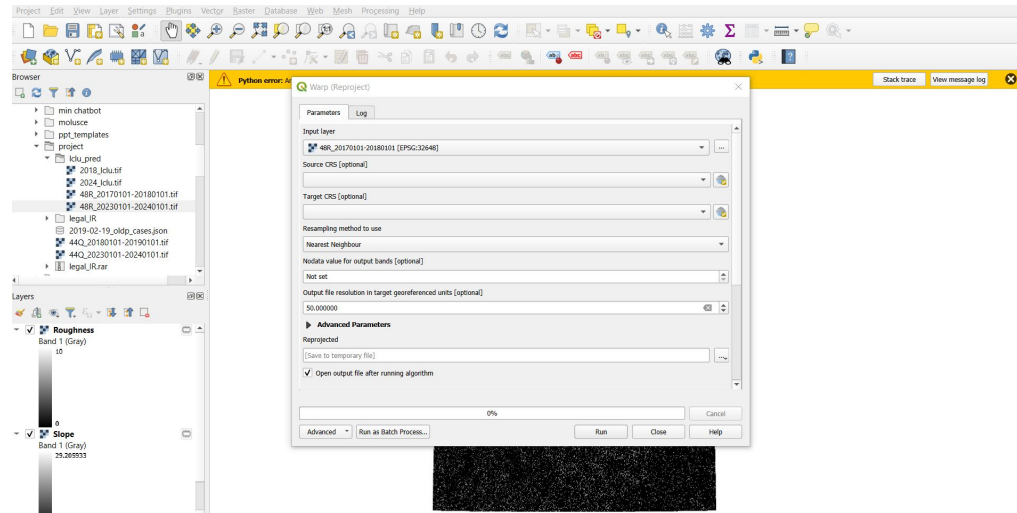
- **Source:** Sentinel-2 satellite imagery.
- **Data Type:** Multi-spectral images covering the area of interest.
- **Land Use Classes:** 9 different land use classes are extracted for analysis.



METHODOLOGY

2. Data Preprocessing

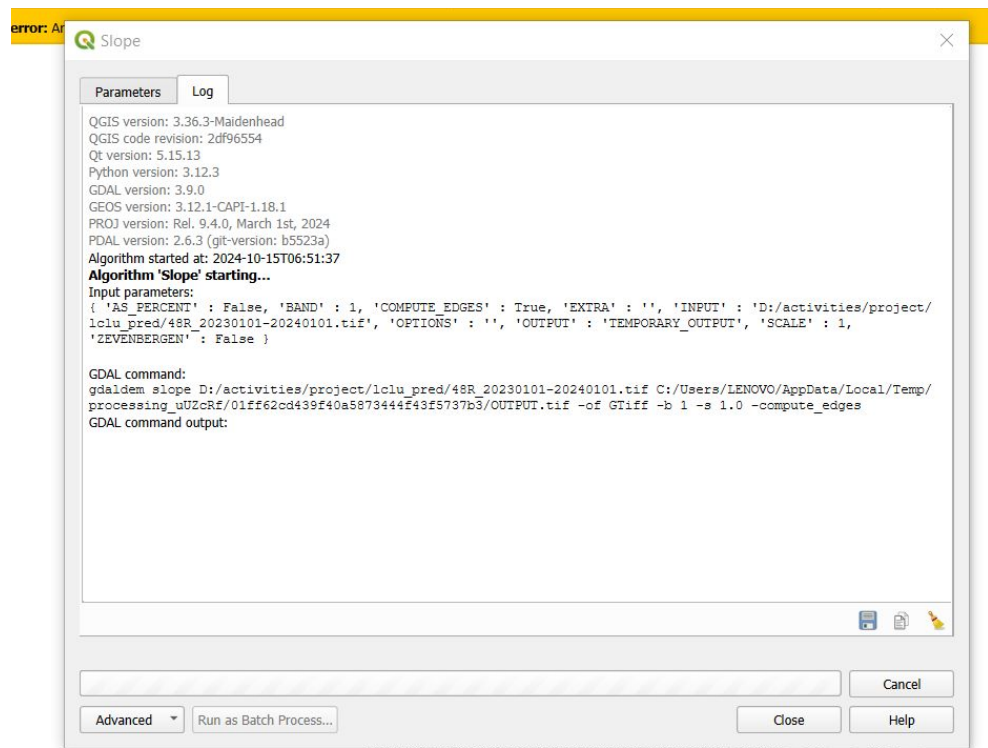
- **Reprojection:** Reproject all images to a common coordinate system to ensure spatial consistency.
- **Resampling:** Resample data to a uniform resolution to standardize pixel sizes across images.
- **Data Cleaning:** Remove noisy or unusable data points (e.g., clouds, shadows) to improve data quality.



METHODOLOGY

3. Feature Extraction

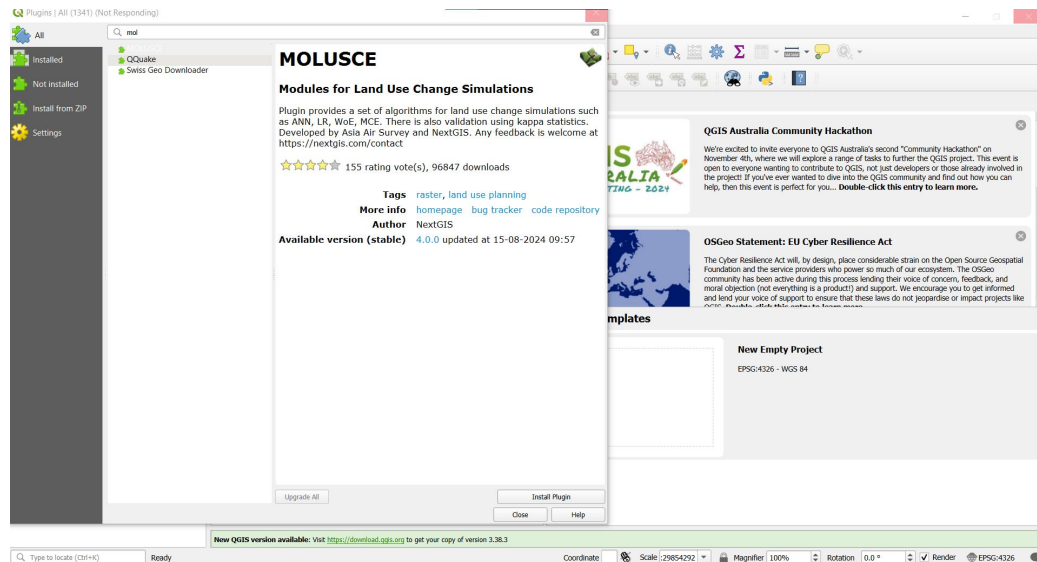
- **Spectral Indices:** Compute relevant spectral indices (NDVI) from multi-spectral bands.
- **Texture Analysis:** Calculate texture features (roughness, slope) to enhance class differentiation.
- **Topographic Indices:** Extract topographic features such as Topographic Position Index (TPI) to understand elevation-based land characteristics.



METHODOLOGY

4. Model Training

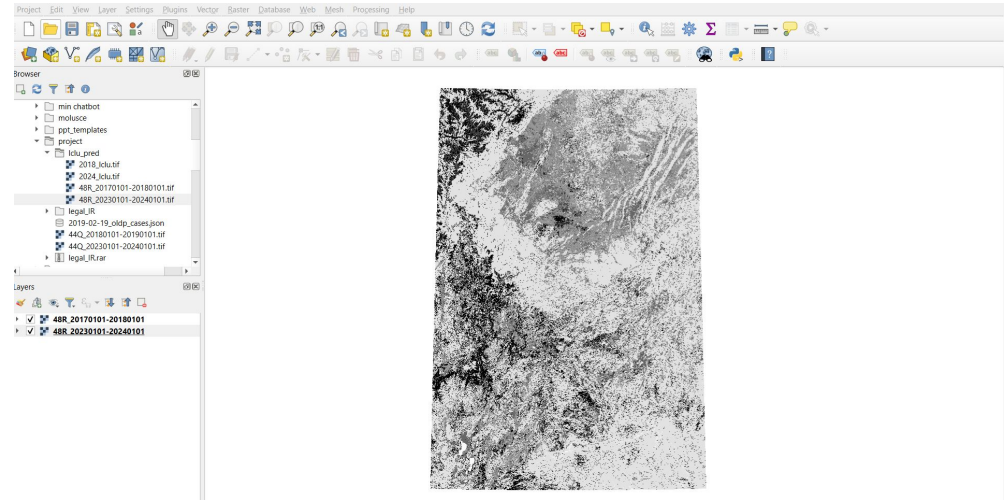
- **Model Selection:** Choose various classification algorithms such as Logistic Regression, Random Forest, ANN, and SVM.
- **Training Set Creation:** Partition the data into training and validation sets to train and evaluate model performance.



METHODOLOGY

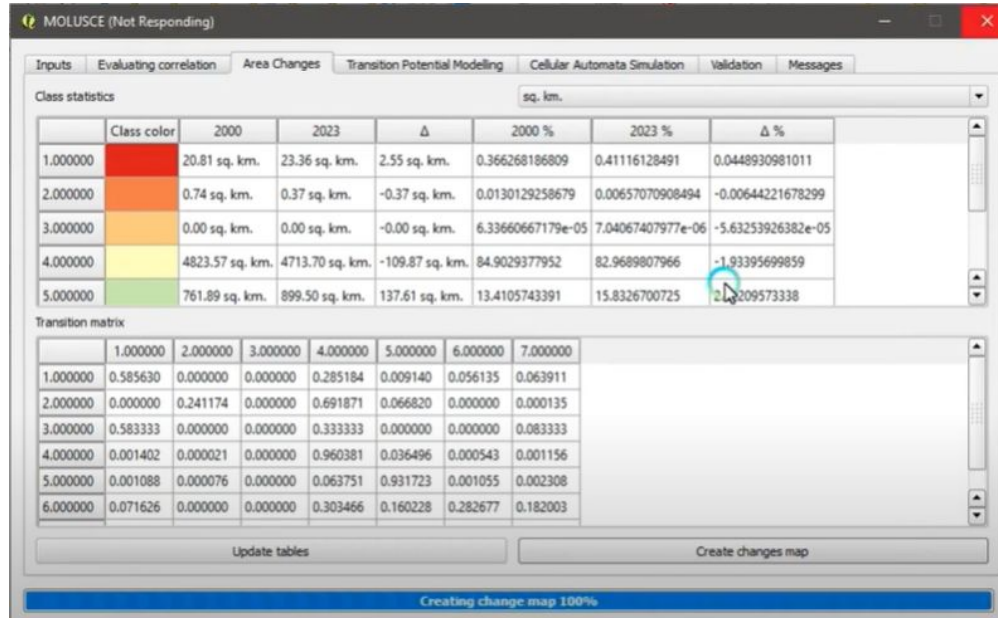
5. Prediction

- **Classification:** Apply trained models to classify the land use classes on test images.
- **Accuracy Assessment:** Compute accuracy metrics (e.g., Overall Accuracy, Kappa Statistic) to evaluate model performance.

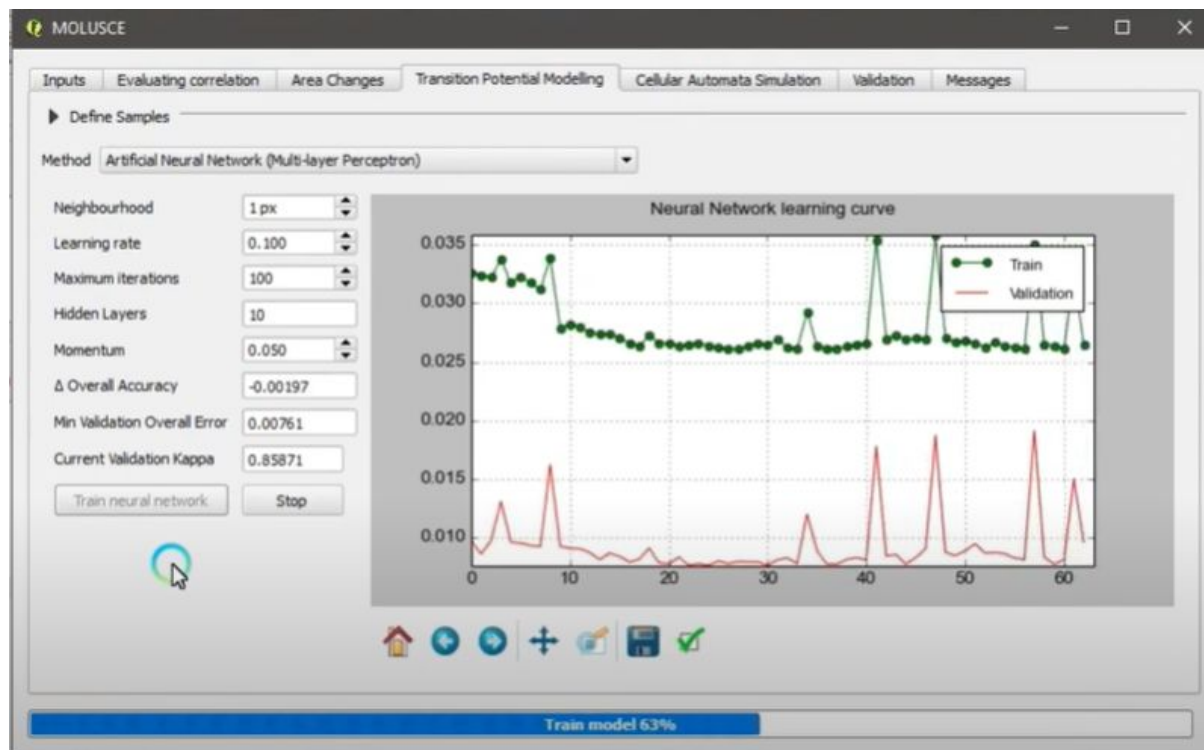


RESULT

Using Sentinel-2 imagery and the MOLUSCE plugin in QGIS, nine land use classes were analyzed to understand land use changes in the Green Valley region from 2017 to 2024. The model's input data consisted of multi-temporal Sentinel-2 satellite images, and changes were projected based on drivers like proximity to urban centers, population density, and economic activity.



RESULT



RESULT

Model	Overall Accuracy	Kappa Statistics
Logistic Regression	86.3 %	0.78
Weight of Evidence	85 %	0.76
Artificial Neural Network	81.6 %	0.68
Multi-Criteria Evaluation	89.7%	0.81

- **Agriculture:** Agricultural areas are expected to decrease by about 15% due to urban sprawl and the conversion of some agricultural lands into forested or barren lands as farming activities shift.
- **Forest:** Forest cover is anticipated to increase by roughly 7%, mainly due to reforestation efforts and conversion from abandoned agricultural land.
- **Water Bodies:** Minimal changes were observed in water bodies, though small encroachments indicate areas potentially at risk from nearby land development.